FIITJEE **JEE (Main)-2018**

ALL INDIA TEST SERIES

FULL TEST – V

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

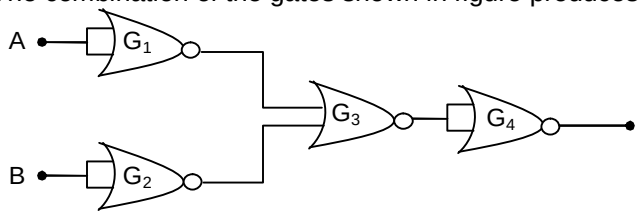
PART – I

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A transmitter supplies 9 kw to the aerial when unmodulated. The power radiated when modulation index is 60%, is
   (A) 9.72 kw (B) 9 kw
   (C) 5.4 kw (D) 10.62 kw

2. The combination of the gates shown in figure produces



   (A) AND gate (B) XOR gate
   (C) NOR gate (D) NAND gate

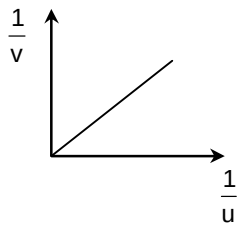
3. Two radioactive materials X_1 and X_2 have decay constants 10λ and λ respectively. If initially they have the same number of nuclei, then the ratio of number of nuclei of X_1 to that of X_2 will be $\frac{1}{e}$ after time
   (A) $\frac{1}{10\lambda}$ (B) $\frac{1}{11\lambda}$
   (C) $\frac{1}{9\lambda}$ (D) $\frac{11}{10\lambda}$

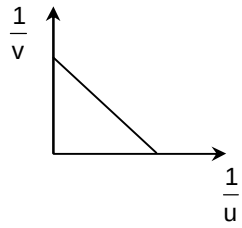
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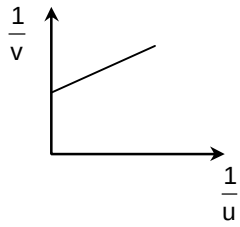
4. When the hydrogen atom is raised from the ground state to fifth state
(A) both K.E and P.E increases (B) both K.E and P.E decreases
(C) P.E increases and K.E decreases (D) P.E decreases and K.E increases

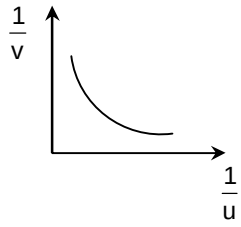
5. In an ideal parallel LC circuit, the capacitors is charged by connecting it to a dc source which is then disconnected. The current in the circuit
(A) becomes zero instantaneously (B) grows monotonically
(C) decays monotonically (D) oscillates instantaneously.

6. Which of the following graph is applicable between $\frac{1}{v}$ and $\frac{1}{u}$ for a concave mirror

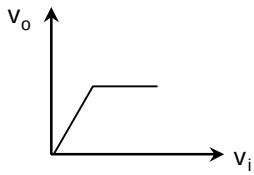
(A) 

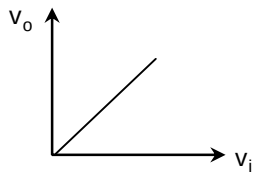
(B) 

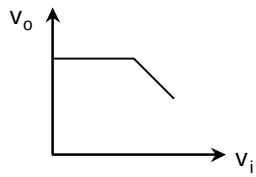
(C) 

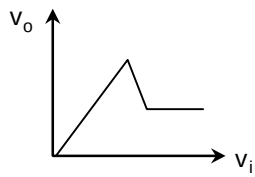
(D) 

7. In a voltage regulating of zener diode, the graph of output voltage v_o versus input voltage v_i is

(A) 

(B) 

(C) 

(D) 

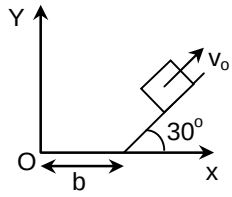
Space for Rough work

8. From ideal instruments, current measured is $i = 10.0\text{amp}$, potential difference measured is V = 100.0 volt, length of wire is 31.4cm, and diameter of wire is 2.00 mm, the resistivity of wire will be (in correct significant figures) $(\pi = 3.14)$

(A) $1.00 \times 10^{-4}\,\Omega - \text{m}$ (B) $1.0 \times 10^{-4}\,\Omega - \text{m}$

(C) $1 \times 10^{-4}\,\Omega - \text{m}$ (D) $1.000 \times 10^{-4}\,\Omega - \text{m}$

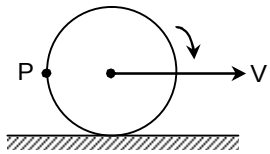
9. A cube of mass m and side a is moving along a plane with constant speed v_o as shown in figure. The magnitude of angular momentum of the cube about z -axis would be.



(A) $\frac{mv_o b}{2}$ (B) $\frac{\sqrt{3}mv_o b}{2}$

(C) $mv_o\left(b - \frac{a}{2}\right)$ (D) None of these

10. A disc of radius R is rolling purely on a flat horizontal surface, with constant angular velocity. The angle between the velocity and acceleration vectors of point P is



(A) zero (B) 45°

(C) 135° (D) $\tan^{-1}\left(\frac{1}{2}\right)$

11. A small mass slides down an inclined plane of inclination θ with the horizontal. The coefficient of friction is $\mu = \mu_o x$ where x is the distance through which the mass slides down and μ_o is constant. Then the distance covered by the mass before it stops is

(A) $\frac{2}{\mu_o}\tan\theta$ (B) $\frac{4}{\mu_o}\tan\theta$

(C) $\frac{1}{2\mu_o}\tan\theta$ (D) $\frac{1}{\mu_o}\tan\theta$

Space for Rough work

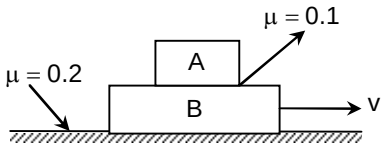
12. Acceleration of a particle moving in x-y plane varies with time t as. $\vec{a} = \left(ti + 3t^2 j\right)$.

Here a is in m/s^2 and t in sec. At time t = 0 particle is at rest at origin. Mass of the particles is 1 kg. Find the net work done on the particle in first 2 sec.

(A) 40 J (B) 34 J

(C) 16 J (D) 48 J

13. Both the blocks as shown in figure are given together a horizontal velocity towards right. The acceleration of the centre of mass of the system of blocks is $\left(m_A = 2m_B = 2kg\right)$.



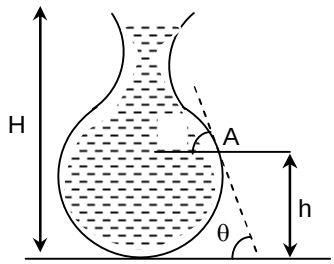
(A) zero

(B) $\frac{5}{3} m/s^2$

(C) $\frac{7}{3} m/s^2$

(D) $2\ m/s^2$

14. Figure here shows the vertical cross section of a vessel filled with a liquid of density ρ. The normal thrust per unit area due to liquid on the walls of the vessel at point A as shown, will be



(A) $h\rho g$

(B) $H\rho g$

(C) $(H-h)\rho g$

(D) $(H-h)\rho g\cos\theta$

15. The volume thermal expansion coefficient of an ideal gas at constant pressure is

(A) T (B) T^2

(C) $\frac{1}{T}$ (D) $\frac{1}{T^2}$

Space for Rough work

16. A heavy but uniform rope of length L is suspended from a ceiling. A particle is dropped from the ceiling at the instant when the bottom end is given a transverse wave pulse. Where will the particle meet the pulse

(A) at a distance $\frac{2L}{3}$ from the bottom (B) at a distance $\frac{L}{3}$ from the bottom

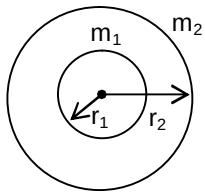
(C) at a distance $\frac{3L}{4}$ from the bottom (D) None of the above

17. A particle moves according to the law $x = a\cos\frac{\pi t}{2}$. The distance covered by it in the time internal between t = 0 to t = 3 sec is.

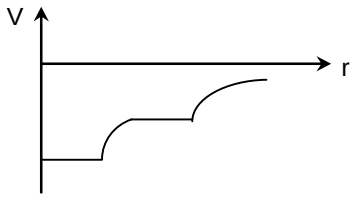
(A) 2a (B) 3a

(C) 4a (D) a

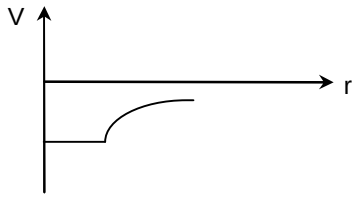
18. Two concentric spherical shells are as shown in figure. The V – r graph will be as



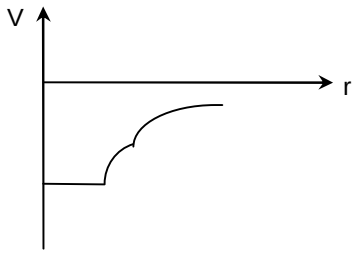
(A)



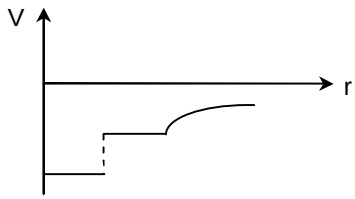
(B)



(C)



(D)



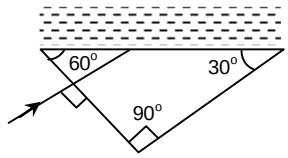
Space for Rough work

19. Two coherent point sources S_1 and S_2 vibrating in phase emit light of wavelength λ. The separation between the sources is 2λ. The smallest distance from S_2 on a line passing through S_2 and perpendicular to S_1S_2, where a minimum of intensity occurs is

(A) $\frac{7\lambda}{12}$ (B) $\frac{15\lambda}{4}$

(C) $\frac{\lambda}{2}$ (D) $\frac{3\lambda}{4}$

20. Light is incident normally on face AB of a prism as shown in figure. A liquid of refractive index μ is placed on face AC of the prism. The prism is made of glass of refractive index $\frac{3}{2}$. The limits of μ for which total internal reflection takes place on face AC is

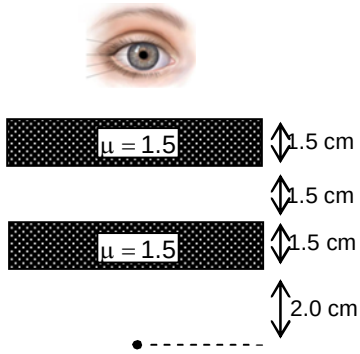


(A) $\mu > \frac{\sqrt{3}}{2}$ (B) $\mu < \frac{3\sqrt{3}}{4}$

(C) $\mu > \sqrt{3}$ (D) $\mu < \frac{\sqrt{3}}{2}$

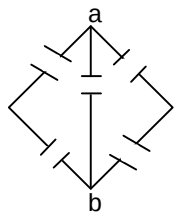
21. The image of point P when viewed from top of the slabs will be

(A) 2 cm above P
(B) 1.5 cm above P
(C) 2 cm below P
(D) 1 cm below P



Space for Rough work

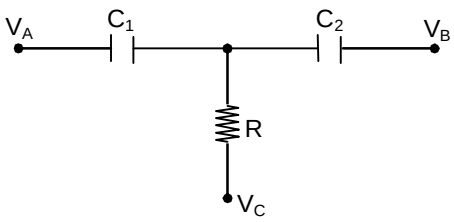
22. If the capacitance of each capacitor is C, then effective capacitance of the shown network across junction ab is



(A) 2 C (B) C

(C) $\frac{C}{2}$ (D) 5 C

23. For given circuit charge on capacitor C, and C_2 in steady state will be equal to



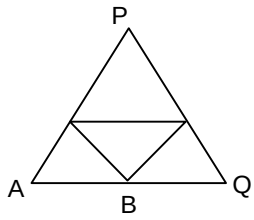
(A) $C_1(V_A - V_C), C_2(V_C - V_B)$ respectively

(B) $C_1(V_A - V_B), C_2(V_A - V_B)$ respectively

(C) $(C_1 + C_2)(V_A - V_B)$ on each capacitor

(D) $\left(\frac{C_1C_2}{C_1 + C_2}\right)(V_A - V_B)$ on each capacitor

24. In the diagram resistance between any two junctions is R. Equivalent resistance across terminals A and B is (Assume P and Q across as a junction)



(A) $\frac{11\,R}{7}$ (B) $\frac{18\,R}{11}$

(C) $\frac{7\,R}{11}$ (D) $\frac{11R}{18}$

25. A particle of charge per unit mass α is released from origin with velocity $\vec{V} = v_0 i$ in a magnetic field. $\vec{B} = -B_0\hat{K}$ for $x \le \frac{\sqrt{3}}{2}\frac{v_0}{B_0\alpha}$ and $\vec{B} = 0$ for $x > \frac{\sqrt{3}}{2}\frac{v_0}{B_0\alpha}$

and $\vec{B} = 0$ for $x > \frac{\sqrt{3}}{2}\frac{v_0}{B_0\alpha}$

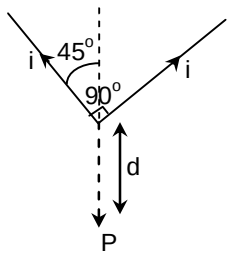
The x – coordinate of the particle at time $t > \left(\frac{\pi}{3B_0\alpha}\right)$ would be

(A) $\frac{\sqrt{3}}{2}\frac{v_0}{B_0\alpha} + \frac{\sqrt{3}}{2}v_0\left(t - \frac{\pi}{B_0\alpha}\right)$ (B) $\frac{\sqrt{3}}{2}\frac{v_0}{B_0\alpha} + v_0\left(t - \frac{\pi}{3B_0\alpha}\right)$

(C) $\frac{\sqrt{3}}{2}\frac{v_0}{B_0\alpha} + \frac{v_0}{2}\left(t - \frac{\pi}{3B_0\alpha}\right)$ (D) $\frac{\sqrt{3}}{2}\frac{v_0}{B_0\alpha} + \frac{v_0 t}{2}$

Space for Rough work

26. Find the magnetic field at P due to the arrangement shown in figure.



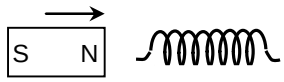
(A) $\frac{\mu_o i}{\sqrt{2}\pi d}\left(1-\frac{1}{\sqrt{2}}\right)\otimes$

(B) $\frac{2\mu_o i}{\sqrt{2}\pi d}\otimes$

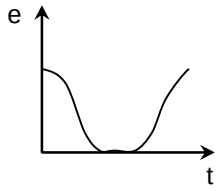
(C) $\frac{\mu_o i}{\sqrt{2}\pi d}\otimes$

(D) $\frac{\mu_o i}{\sqrt{2}\pi d}\left(1+\frac{1}{\sqrt{2}}\right)\otimes$

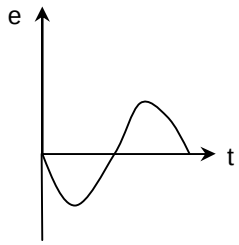
27. The variation of induced emf(e) with time (t) in a coil if a short bar magnet is along its axis with a constant velocity is best represented as



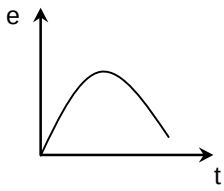
(A)



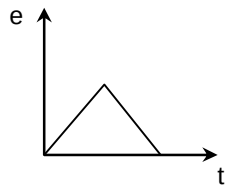
(B)



(C)

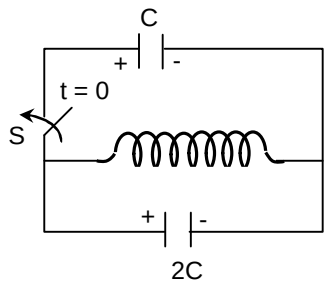


(D)



Space for Rough work

28. In the given LC circuit if initially capacitor C has charge Q on it and 2C has charge 2Q. The polarities are as shown in the figure. Then after closing the switch s at t = 0



(A) energy will get equally distributed in both the capacitor just after closing the switch

(B) Initial rate of growth of current in inductor will be $2Q / 3CL$

(C) Maximum energy in the inductor will be $3Q^2 / 2C$

(D) None of these

29. In an insulator, the forbidden energy gap between the balance band and conduction band is of the order of

(A) 1 MeV (B) 0.1 MeV

(C) 1 eV (D) 5 eV

30. A TV tower has a height of 150 m. The area of the region covered by the TV broadcast is (Radius of earth = $6.4 \times 10^6 m$)

(A) $9.6\pi \times 10^8 m^2$ (B) $19.2\pi \times 10^8 m^2$

(C) $19.2\pi \times 10^{10} m^2$ (D) $1.92 \times 10^3 km^2$

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. An aqueous solution has a density of 1.37 g/ml. If molecular wt of solute is 68.5 g/mol, then calculate molarity of solution.
(A) 2 M (B) 50 M
(C) 5 M (D) 20 M

32. Calculate the degree of ionisation of 0.04 M HOCl solution having ionisation constant 1.25×10^{-4} ?
(A) 0.025 (B) 0.5
(C) 0.25 (D) 0.055

33. In a crystalline solid, X atoms occupy hcp and ¼ th of tetrahedral voids are occupied by Y atoms. Then empirical formula of crystalline solid is
(A) X_2Y (B) XY_2
(C) X_4Y (D) XY

34. Bohr's model is able to explain
(A) Zeeman effect (B) Stark effect
(C) Bond formation of H_2 (D) Hydrogens spectrum

35. On increasing temperature from 200 K to 220 K rate of reaction A increase by 3 times and rate of reaction B increases by 9 times then correct relationship between activation energy of A and B is
(A) $E_A = 3\ E_B$ (B) $3\ E_A = E_B$
(C) $E_B = 2E_A$ (D) $E_A = 2E_B$

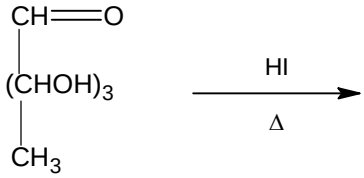
36. Calculate work done during isothermal reversible process when 5 mol ideal gas is expanded so that its volume is doubled at 400 K.
(A) – 11.5 kJ (B) – 344 kJ
(C) 0 (D) – 2.8 kJ

Space for Rough work

37. The K_{sp} of AgCl at 25°C is 2.56×10^{-10}. Then how much volume of H_2O is required to dissolve 0.01 mole of salt?
(A) 800 L
(B) 400 L
(C) 625 L
(D) 50 L

38. Aqua regia convert
(A) Pt to $[PtCl_4]^{-2}$
(B) Pt to $PtCl_2$
(C) Ag to AgCl
(D) Au to $[AuCl_4]^-$

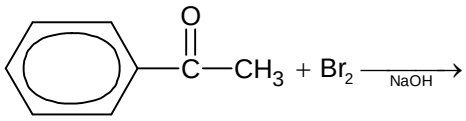
39. Identify product of the reaction

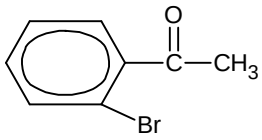


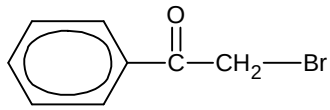
(A) n – Pentane
(B) 2 – Pentane
(C) 1, 3 Pentadiene
(D) 3 - Pentene

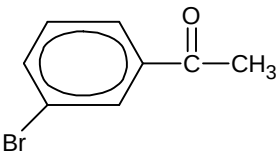
40. In extraction of a metal by bessemerization, Cu_2S is converted into
(A) Cu_2O
(B) CuO
(C) Cu
(D) $CuFeO_2$

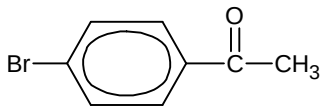
41.



(A)


(B)


(C)


(D)


Space for Rough work

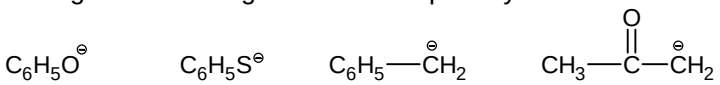
42. Arrange the given compounds is order of their basic strength

(i) Histidine (ii) Lysine (iii) Arginine

(A) i > ii > iii (B) ii > i > iii

(C) iii > ii > i (D) i > iii > ii

43. Arrange the following in their nucleophilicity order.

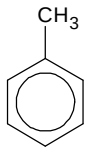


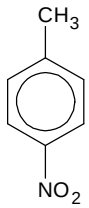
(I) (II) (III) (IV)

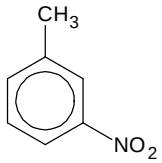
(A) II > III > IV > I (B) I > IV > II > III

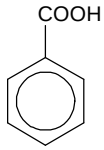
(C) II > IV > III > I (D) III > IV > I > II

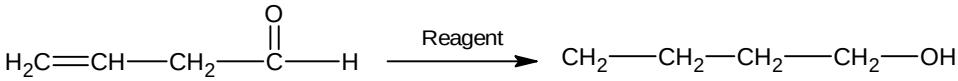
44. The best starting substrate to prepare 2-Bromobenzoic acid is

(A) 

(B) 

(C) 

(D) 

45. 

Appropriate reagent for above conversion will be

(A) Metal + H_2 (B) $LiAlH_4$

(C) $NaBH_4$ (D) All

Space for Rough work

46. An octahedral complex having formula $Ma_2b_2C_2$, which of the following combination is possible to exhibit optical isomerism

(A) $a = Cl^-$, B & C = (en)

(B) a & b & c = EDTA

(C) a & $b = C_2O_4^{-2}$,c = en

(D) $a = H_2O$, $b = NH_3$, $C = Cl^-$

47. Which of the following statement is incorrect?

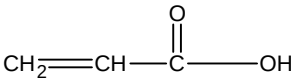
(A) It is possible to separate eclipsed and staggered form of ethane at room temperature

(B) Eclipsed form has torsional strain

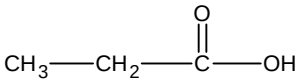
(C) Ethane has infinite number of conformers

(D) Energy difference between staggered form & eclipsed form of ethane is 12.5 kJ/mole

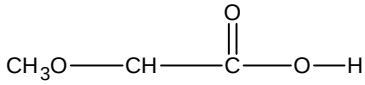
48. Arrange the following compounds in their acidic strength order.



(I)



(II)



(III)

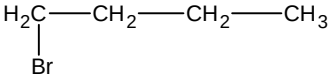
(A) I > II > III

(B) III > II > I

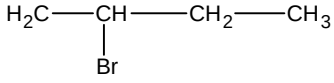
(C) I > III > II

(D) II > I > III

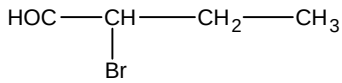
49. Arrange the following compound in order of reactivity towards SN^1 reaction.



(I)



(II)



(III)

(A) I > II > III

(B) II > I > III

(C) III > II > I

(D) I > III > II

50. Which of the following statement is not true regarding B_2H_6?

(A) It reacts with NaH to form $NaBH_4$

(B) It is inflammable gas

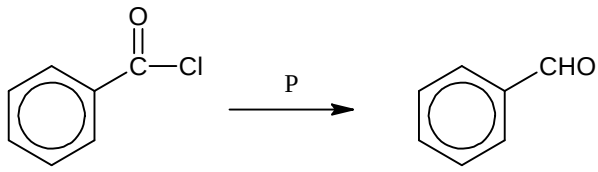
(C) It contain four equal B – H bonds

(D) it reacts with Lewis base always to form always adduct

Space for Rough work

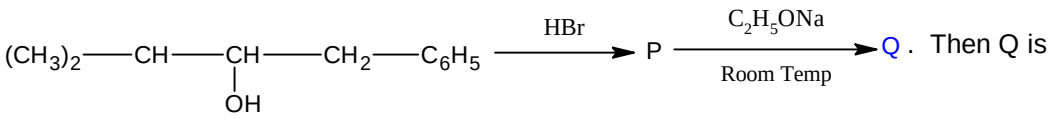
51. Regarding brown ring complex formation, which of the following statement is wrong?
(A) NO formation takes place
(B) $Fe_2(SO_4)_3$ is used in this process
(C) Fe^{+2} and NO_3^- both are reduced
(D) Formula of brown ring complex is $\left[Fe(H_2O)_5NO\right]^{+2}$

52. Identify incorrect statement regarding glucose.
(A) Pentaacetate of glucose does not react with hydroxyl amine
(B) Glucose forms cyclic hemiacetal
(C) Glucose exists in two crystalline forms
(D) α - form of glucose has higher melting point than its β-form

53. 

P will be
(A) $Pd/BaSO_4 + H_2$
(B) $LiAlH_4$
(C) $NaBH_4$
(D) $Pd + H_2$

54. Amoxycillin is
(A) Broad spectrum antibiotic
(B) Bacteriostatic
(C) Synthetic modification of penicillin
(D) (a) & (c) both

55. 

. Then Q is
(A) $(CH_3)_2CH-CH=CH-C_6H_5$
(B) $(CH_3)_2CH-CH(OC_2H_5)-CH_2-C_6H_5$
(C) $(CH_3)_2CH-CH_2-CH(OC_2H_5)-C_6H_5$
(D) $CH_2=C(CH_3)-CH_2-CH_2-C_6H_5$

Space for Rough work

56. If radius of octahedral void is r and radius of atoms which shows close packing is R, then what will be radius of void in terms of radius of atoms?

(A) $\left(\sqrt{2}-1\right)R$ (B) $\sqrt{2}R$

(C) $\frac{R}{\sqrt{2}}$ (D) $\frac{R}{2\sqrt{2}}$

57. For a first order reaction, rate constant is given is $\log k = 14 - \frac{1.2\times10^4}{T}$, then what will be value of temperature if its half life period is 6.93×10^{-3} min.?

(A) 100 K (B) 1000 K

(C) 720 K (D) 327 K

58. What will be solubility of A_2X_3 if its solubility product (K_{sp}) is equal to 1.08×10^{-23} ?

(A) 10^{-5} mol/L (B) 3.7×10^{-4} mol/L

(C) 1.2×10^{-3} mol/L (D) 7.5×10^{-4} mol/L

59. Which of the following is correct for repulsive process in real gas?

(A) Z > 1; repulsive forces are dominating (B) Z < 1; repulsive forces are dominating

(C) short range interaction (D) long range interaction

60. In solid phase, XeF_6 consists of

(A) $XeF_5^+ + F^-$ (B) $XeF_4 + F_2$

(C) $XeF_5^+ + XeF_2^-$ (D) $XeF_2 + 2F_2$

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If α,β,γ are roots of equation $x^3-3x^2+3x+7=0$, then $\sum\frac{\alpha-1}{\beta-1}$ is

(A) 0 (B) 3ω

(C) $\frac{3}{\omega}$ (D) $2\omega^2$

62. If (x, y) satisfies the equation $x^2-xy+y^2=4(x+y-4)$, then value of $\frac{x}{y}+\frac{y}{x}$ is $(x,y\in R)$

(A) 1 (B) 2

(C) 4 (D) 0

63. Let ab^2c^3, $a^2b^3c^4$, $a^3b^4c^5$ are in A.P. (a, b, c > 0), then minimum value of $27a+8b+125c$ is

(A) 30 (B) 60

(C) 90 (D) 120

64. If $f(x)$ is not identically zero and $\int_{-5}^{5}f(x)dx=0$, then value of $\int_{-2}^{3}f(x)\left[\frac{1}{x^2+1}\right]dx$ equals to ([.] is G.I.F.)

(A) 0 (B) 1

(C) Depends on $f(x)$ (D) None of these

65. Value of $\lim_{n\to\infty}\frac{1}{(n^3+1)}+\frac{2^2}{(n^3+2^3)}+.....+\frac{1}{2n}$ is

(A) $\ln 2$ (B) $\frac{1}{2}\ln 3$

(C) $\frac{1}{4}\ln 3$ (D) $\frac{1}{3}\ln 2$

Space for Rough work

66. Total number of words using all the letters of word INDIA in which A doesn't come in middle is

(A) 36 (B) 48

(C) 60 (D) 12

67. If a, b, c are in A.P. and sum of coefficients of $\left(1+\left(ax^2-2bx+c\right)^2\right)^{2017}$ is equal to $-y^2-y$, then value of $y^{2018}+\frac{1}{y^{2018}}+1$ is

(A) 0 (B) 1

(C) 2 (D) None of these

68. In a triangle ABC, the value of $\Delta=\begin{vmatrix}\sin(A+B+C) & \sin B & \cos C\\ -\sin B & 0 & \tan A\\ \cos(A+B) & -\tan A & 0\end{vmatrix}$ is equal to

(A) –1 (B) 0

(C) 1 (D) 2

69. Value of $\int_{-3}^{3}\frac{x^2}{\left(1+|x|^3\right)\left(3^x+1\right)}dx$ is

(A) $3\ln 2$ (B) $\ln 3$

(C) $2\ln 3$ (D) None of these

70. Let f be a differential function satisfying $f(x)+f(x+1)+.....+f(x+99)=50\ \forall x\in R$, $f(0)=\frac{1}{3}$, then the value of $\lim_{x\to\infty}\left(f\left(\frac{1}{x}\right)+f^2\left(\frac{1}{x}+100\right)+f^3\left(\frac{1}{x}+200\right)+.....\infty\right)$ is

(A) $\frac{1}{2}$ (B) 1

(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

Space for Rough work

71. If $P(B)=\frac{3}{4}$, $P\left(A\cap B\cap \bar{C}\right)=\frac{1}{3}$, $P\left(\bar{A}\cap B\cap \bar{C}\right)=\frac{1}{3}$, then $P(B\cap C)$ is

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$

(C) $\frac{1}{12}$ (D) $\frac{1}{9}$

72. Let A and B are two sets, then $(A\cup B)'\cup(A'\cap B)$ is equal to

(A) A' (B) A

(C) B' (D) None of these

73. If the relation $R: A\to B$, where A = {1, 2, 3, 4} and B = {1, 3, 5} is defined by $R=\{(x, y): x<y,\ x\in A,\ y\in B\}$, then $R^{-1}OR$ is

(A) $\{(1, 3)(1, 5)(2, 3)(2, 5)(3, 5)(4, 5)\}$ (B) $\{(3, 1)(5, 1)(5, 2)(5, 3)(5, 4)\}$

(C) $\{(3, 3)(3, 5)(5, 3)(5, 5)\}$ (D) None of these

74. If p, q, r are statements with truth values false, true and false respectively, then truth value of $(\sim p\vee\sim q)\vee r$ is

(A) True (B) False

(C) False, if r is true (D) False, if q is false

75. For a series if n = 10, $\sum x=60$ and $\sum x^2=1000$, then standard deviation is

(A) 8 (B) 64

(C) 24 (D) 128

76. If $1+\cot\theta\le\cot\frac{\theta}{2}$ for $0<\theta<\pi$, then θ is equal to

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{8}$

Space for Rough work

77. In a triangle ABC, if b = 2, $c = \sqrt{3}$, $\angle A = 30^o$, then area of incircle of triangle is

(A) $\pi\left(2-\frac{\sqrt{3}}{2}\right)$ (B) $\pi\left(1-\frac{\sqrt{3}}{2}\right)$

(C) $\sqrt{2}\pi$ (D) $\frac{\pi\sqrt{3}}{2}$

78. From the top of a cliff 50 m high, the angle of depressions of the top and bottom of a tower are observed to be 30° and 45°, then the height of tower is

(A) 50 m (B) $50\sqrt{3}$ m

(C) $50\left(\sqrt{3}-1\right)$ m (D) $50\left(1-\frac{\sqrt{3}}{3}\right)$ m

79. The matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ -1 & -2 & -3 \end{bmatrix}$ is

(A) Idempotent (B) Nilpotent

(C) Involutary (D) Orthogonal

80. Equation of plane perpendicular to x = 0 and passing through (2, –1, 0) can be

(A) $x+3y+z+1=0$ (B) $2y-z+2=0$

(C) $y=-1$ (D) $2y+z+1=0$

81. The area of triangle formed by positive y-axis and the normal and tangent to $x^2+y^2=4$ at $\left(1, \sqrt{3}\right)$ is

(A) $\sqrt{3}$ sq. unit (B) $\frac{2}{\sqrt{3}}$ sq. unit

(C) $2\sqrt{3}$ sq. unit (D) $4\sqrt{3}$ sq. unit

82. Equation of tangent to $y=\int_0^{x^2}\frac{t^2-1}{t^2+1}dt$ at x = 1 is

(A) x = 0 (B) $x=\frac{1}{\pi}-1$

(C) $y=1-\pi$ (D) $y=1-\frac{\pi}{2}$

Space for Rough work

83. If $f(x) = |x-1| + |x-3| + |5-x|$ is symmetrical about the line $x = 1 + |\lambda|$, then λ can be

(A) 3 (B) –2
(C) –1 (D) 1

84. $\int \frac{e^{\tan\theta}\left(\tan^3\theta + 3\tan^2\theta + \tan\theta\right)}{1+\tan^2\theta} d\theta$ is equal to

(A) $e^{\tan\theta}\left(\tan^3\theta + \tan\theta\right) + c$ (B) $e^{\tan\theta}\left(\tan^3\theta + \tan^2\theta\right) - e^{\tan\theta} + c$
(C) $e^{\tan\theta}\left(\tan\theta\sec^2\theta - 1\right) + c$ (D) None of these

85. Solution of $y^5 x + y - x\frac{dy}{dx} = 0$ is

(A) $\frac{y^5}{5} + \frac{y^4}{4x^4} = c$ (B) $\frac{x^4}{4} + \frac{x^5}{5y^5} = c$
(C) $\frac{y^4}{4} + \frac{y^5}{5x^4} = c$ (D) $\frac{x^5}{5} + \frac{x^4}{4y^4} = c$

86. The points on the axis of parabola $3y^2 + 4y - 6x + 8 = 0$, from where 3 distinct normal can be drawn is given by

(A) $\left(\lambda, \frac{4}{3}\right)$, where $\lambda > \frac{19}{9}$ (B) $\left(\lambda, \frac{-2}{3}\right)$, where $\lambda > \frac{19}{9}$
(C) $\left(\lambda, \frac{1}{3}\right)$, where $\lambda > \frac{7}{9}$ (D) None of these

87. If (a, b) is a point on the ellipse $x^2 + 2y^2 = 6$ which is at a minimum distance from the line $x + y = 7$, then $a^2 + b^2$ is

(A) 4 (B) 5
(C) 6 (D) 8

Space for Rough work

88. If e is the eccentricity of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and θ is angle between the asymptote, then $\cos\frac{\theta}{2}$ is equal to

(A) $\frac{1-e}{e}$ (B) $\frac{1}{e}$

(C) $\frac{1}{e}+1$ (D) None of these

89. The roots of $(z+\alpha\beta)^3=\alpha^3$ $(\alpha\neq 0)$ represent the vertices of triangle of the side length

(A) $\frac{1}{\sqrt{3}}(\alpha\beta)$ (B) $\sqrt{3}|\alpha|$

(C) $\sqrt{3}|\beta|$ (D) $\frac{|\alpha|}{\sqrt{3}}$

90. A set has $(2n+1)$ elements, then total number of non empty subsets of given set, containing at most n elements is

(A) 2^{2n-1} (B) 2^{2n}

(C) $2^{2n}-1$ (D) $2^{2n+1}-1$

Space for Rough work